Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

Data Visualization with Power BI and Tableau

Practical – IX

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Class: TYIT	Batch:01
Date of Assignment: 27-08-2026	Date/Time of Submission:27-08-2026

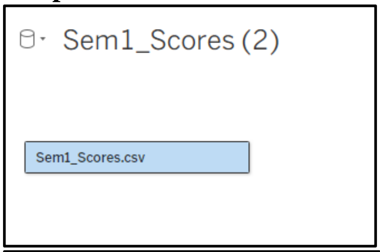
Cleaning and Structuring Data for Analysis

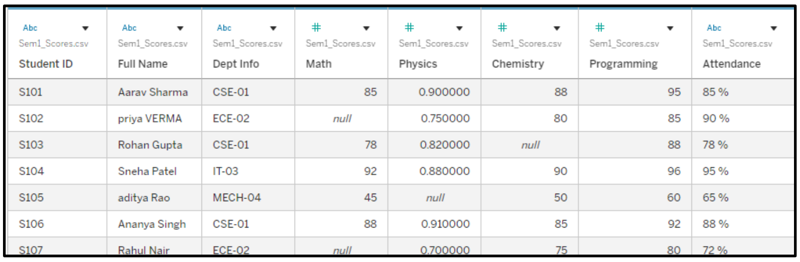
A) Identify and handle missing or inconsistent data.

Steps:-

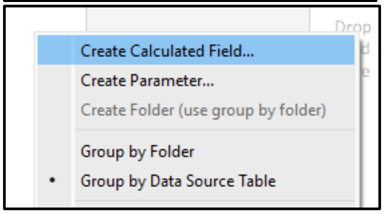
1. Open **Tableau Desktop**, load the given dataset and go to **Sheet 1**.
2. Right-click in the **Data Pane → Create Calculated Field**, name it **Physics_Clean**, and enter: **FLOAT(REPLACE(STR([Physics]), '%', ''));** similarly create **Attendance_Clean** using **FLOAT(TRIM(REPLACE(STR([Attendance]), '%', ''))).**
3. Create calculated fields Math_Clean and Chemistry_Clean using the formulas: **IFNULL([Math], 0)** and **IFNULL([Chemistry], 0).**
4. Edit **Physics_Clean** and use: **IFNULL(FLOAT(REPLACE(STR([Physics]), '%', '')), 0)** to handle missing Physics values.
5. Click **OK**, verify the cleaned fields in the Data Pane, and use the cleaned data for analysis.

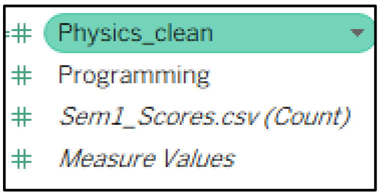
Output:-

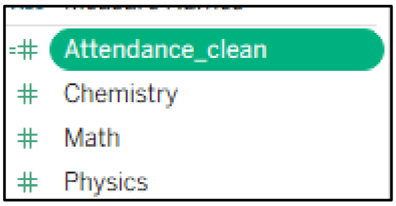


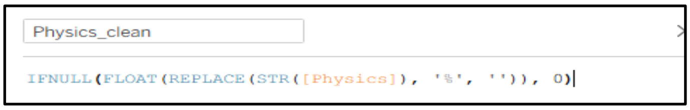


Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %
S102	priya VERMA	ECE-02	null	0.750000	80	85	90 %
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %
S105	aditya Rao	MECH-04	45	null	50	60	65 %
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72 %









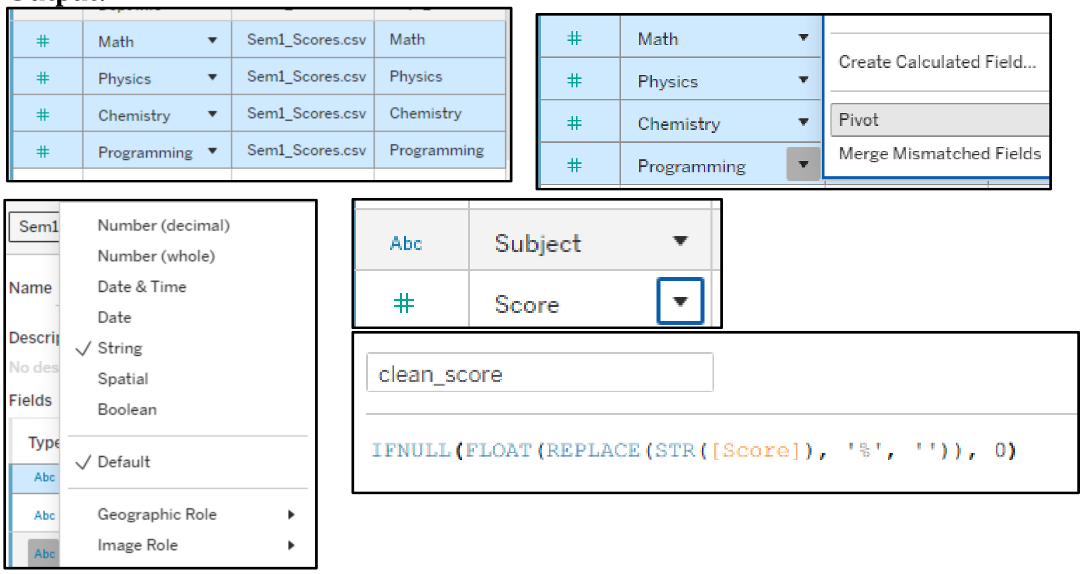
B)Convert wide data into tall format.

Steps:-

1. Open the **Data Source** tab in Tableau and select the required dataset.
2. Hold **Ctrl** and select **Math, Physics, Chemistry,** and **Programming** columns.
3. Right-click on the selected columns and choose **Pivot** to convert the wide data into tall format.
4. Rename **Pivot Field Names** as **Subject** and **Pivot Field Values** as Score.

5. Create a calculated field **Clean_Score** using **IFNULL(FLOAT(REPLACE(STR([Score]), '%', '')), 0)** and verify the converted data.

Output:-

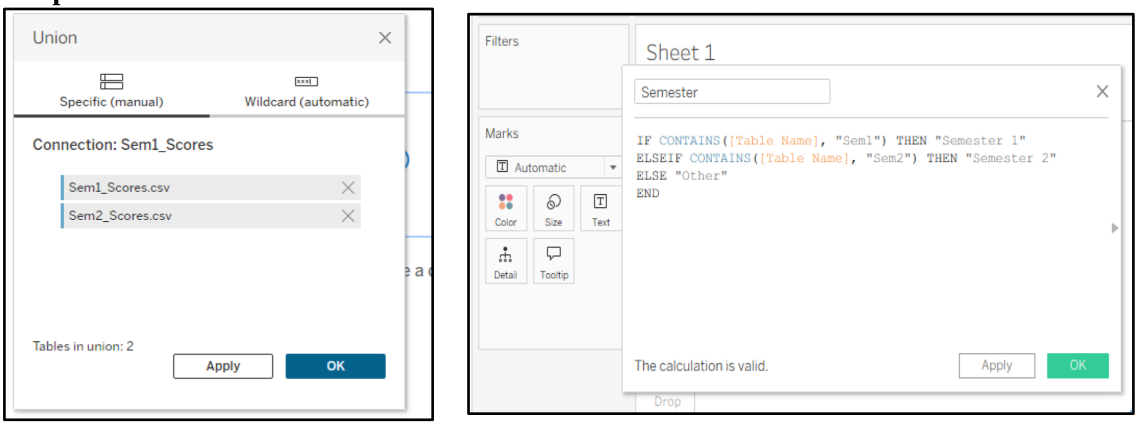


C) Perform union of multiple files.

Steps:

1. Open the Data Source tab and clear the existing canvas if required.
2. Drag New Union onto the canvas and add **Sem1_Scores.csv** and **Sem2_Scores.csv.**
3. Check the data grid at the bottom and verify that the combined dataset contains 20 rows.
4. Scroll to the Table Name column and verify that the first 10 rows belong to **Sem1_Scores.csv** and the remaining 10 rows belong to **Sem2_Scores.csv.**
5. Verify the merged dataset and click Sheet 1 to use the Union data for analysis.

Output:-

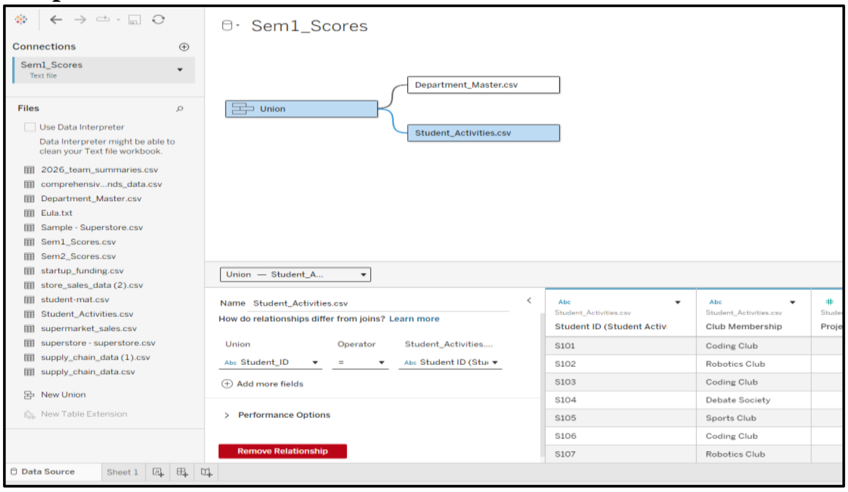


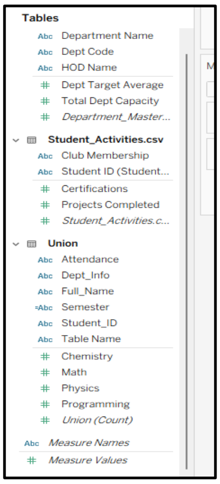
D) Execute cross-database joins.

Steps:

1. Stay on the Data Source canvas and keep the Union table on the canvas.
2. Drag `Department_Master.csv` to the right of the Union table and connect `Dept_Info` with `Dept_Code` using the `=` operator.
3. Drag **`Student_Activities.csv`** onto the canvas and connect **`Student_ID`** from the Union table with **`Student_ID`** from Student_Activities using `=`.
4. Click Sheet 1 and check the Data Pane for fields from Union, Department_Master, and Student_Activities.
5. Verify that the related fields are available and use the combined data for further analysis and visualization.

Output:-



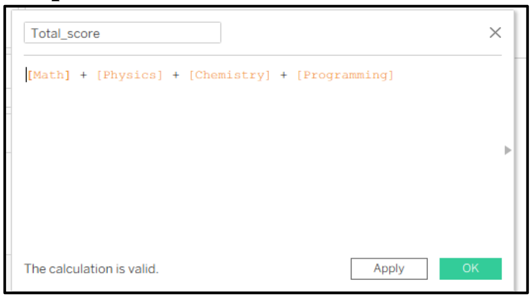


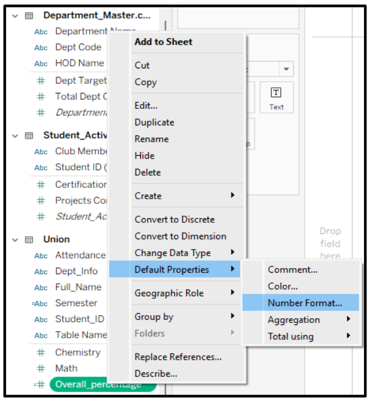
E) Restructure data for better visualization.

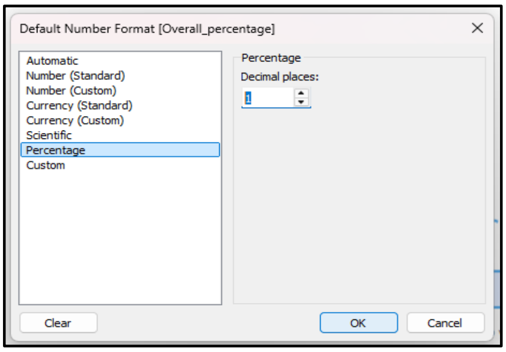
Steps:

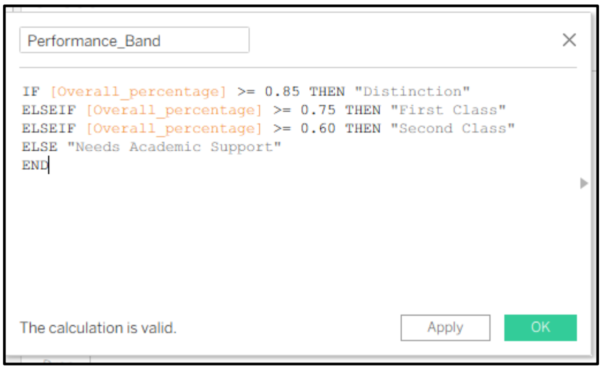
1. Open Sheet 1, create a calculated field named **Total_Score**, and enter: **[Math_Clean] + [Physics_Clean] + [Chemistry_Clean] + [Programming].**
2. Create **Overall_Percentage** using: **[Total_Score] / 400.0**, then format it as Percentage with 1 decimal place.
3. Create Performance_Band using: **IF [Overall_Percentage] >= 0.85 THEN "Distinction" ELSEIF [Overall_Percentage] >= 0.75 THEN "First Class" ELSEIF [Overall_Percentage] >= 0.60 THEN "Second Class" ELSE "Needs Academic Support" END.**
4. Create Attendance_Status using: **IF [Attendance_Clean] >= 75 THEN "Eligible (Good Standing)" ELSE "Shortage Warning (<75%)" END.**
5. Build the scorecard by placing Student_ID and Full_Name on Rows, Semester on Columns, Overall_Percentage on Text, and Performance_Band on Color.

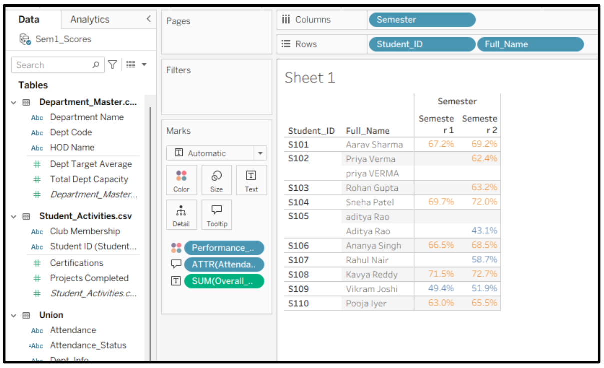
Output:-









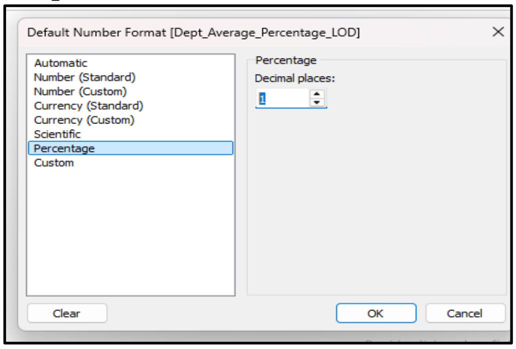


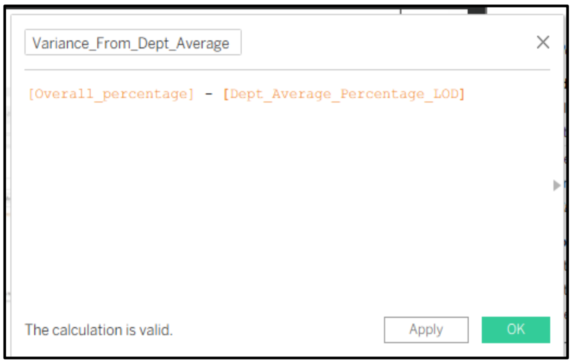
F) Handle different levels of detail in datasets

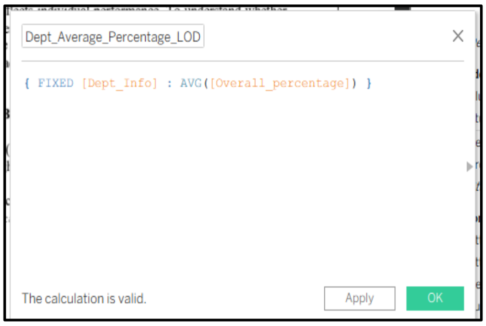
Steps:

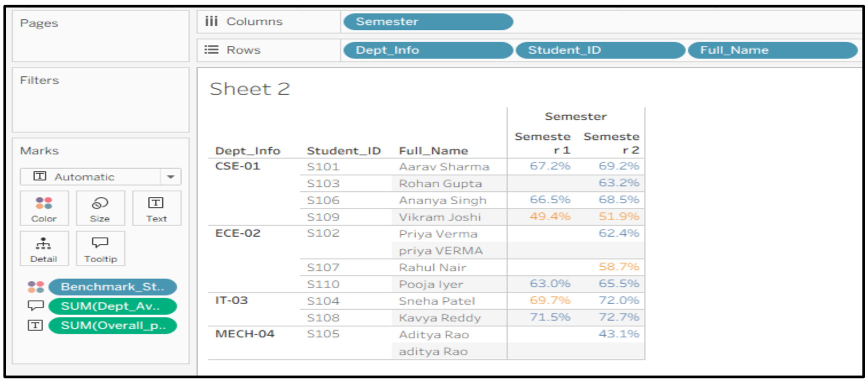
1. Create Dept_Average_Percentage_LOD using: { FIXED [Dept_Info] : AVG([Overall_Percentage]) }.
2. Create Student_Overall_Career_Average using: { FIXED [Student_ID] : AVG([Overall_Percentage]) }.
3. Create Variance_From_Dept_Average using: [Overall_Percentage] - [Dept_Average_Percentage_LOD].
4. Create Benchmark_Status using: IF [Variance_From_Dept_Average] >= 0 THEN "Above Department Benchmark" ELSE "Below Department Benchmark" END.
5. Create the LOD visualization by placing Dept_Info, Student_ID, and Full_Name on Rows, Semester on Columns, Overall_Percentage on Text, Dept_Average_Percentage_LOD on Tooltip, and Benchmark_Status on Color.

Output:-





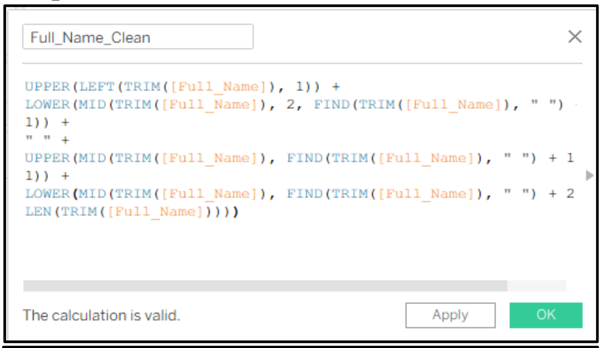


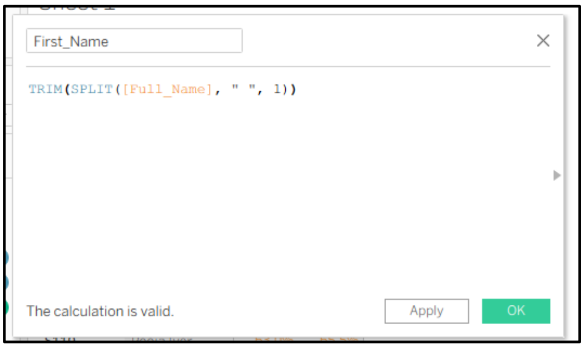


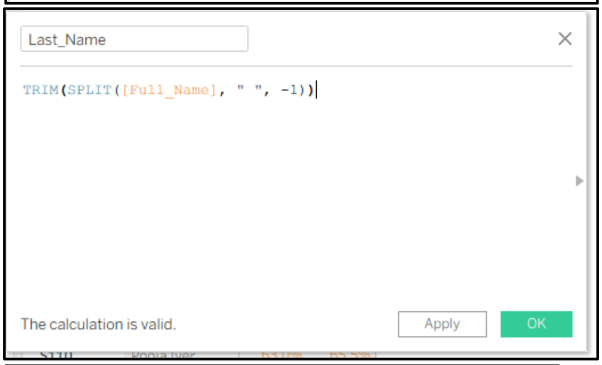
G) Apply advanced data cleaning techniques.
Steps:

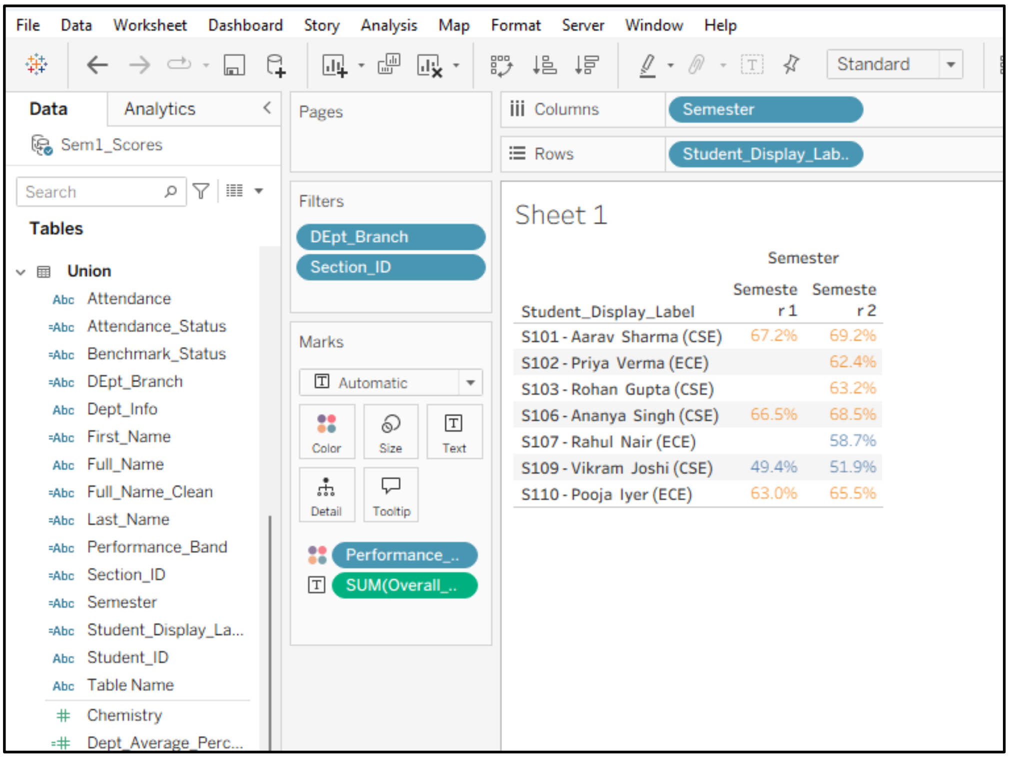
1. Create a calculated field named Full_Name_Clean and apply the name-cleaning formula using TRIM, UPPER, LOWER, LEFT, MID, FIND, and LEN functions.
2. Create First_Name using: TRIM(SPLIT([Full_Name], " ", 1)) and create Last_Name using: TRIM(SPLIT([Full_Name], " ", -1)).
3. Create Dept_Branch using: SPLIT([Dept_Info], "-", 1) and Section_ID using: SPLIT([Dept_Info], "-", 2).
4. Create Student_Display_Label using: [Student_ID] + " - " + [Full_Name_Clean] + " (" + [Dept_Branch] + ")".
5. Click OK, verify all calculated fields in the Data Pane, and use the cleaned fields for visualization.

Output:-








File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Standard
Data
Analytics
Sem1_Scores
Search
Tables
Union
Attendance
Attendance_Status
Benchmark_Status
DEpt_Branch
Dept_Info
First_Name
Full_Name
Full_Name_Clean
Last_Name
Performance_Band
Section_ID
Semester
Student_Display_La...
Student_ID
Table Name
Chemistry
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Pages
Filters
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Size
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Tooltip
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Columns
Semester
Rows
Student_Display_Lab..
Sheet 1
Semester
Student_Display_Label
Semester 1
Semester 2
S101 - Aarav Sharma (CSE) 67.2% 69.2%
S102 - Priya Verma (ECE) 62.4%
S103 - Rohan Gupta (CSE) 63.2%
S106 - Ananya Singh (CSE) 66.5% 68.5%
S107 - Rahul Nair (ECE) 58.7%
S109 - Vikram Joshi (CSE) 49.4% 51.9%
S110 - Pooja Iyer (ECE) 63.0% 65.5%